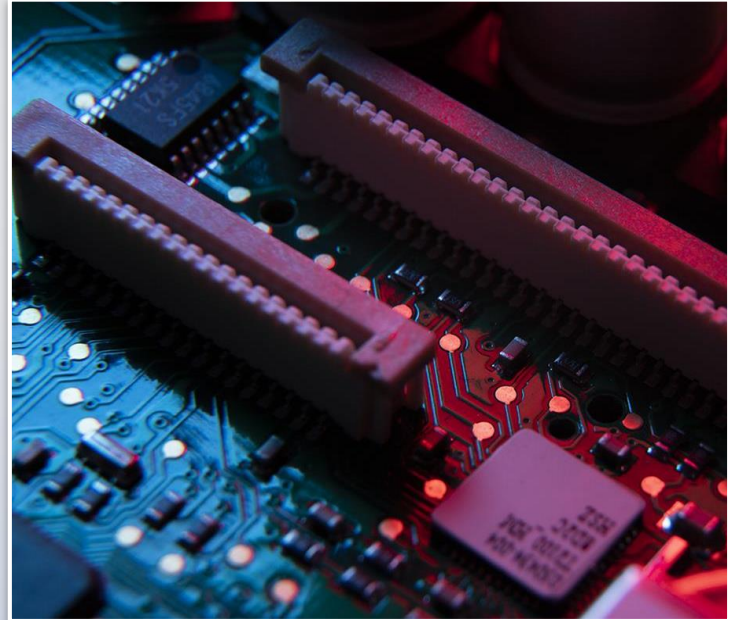


ARDUINO BASED INDUSTRIAL SAFETY SYSTEM

L&T COURSE – MICROCONTROLLERS AND
ITS APPLICATIONS

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ABSTRACT

The primary goal of this industrial safety project is to mitigate the impact of fire incidents in industrial settings, particularly those caused by the leakage of petroleum, chemicals, and kerosene oil, which can lead to both human casualties and property damage. It is crucial to implement a system in every industrial location to enhance security and promptly issue alerts in case of emergencies. This system can also relay critical information to the Occupational Safety and Health (OSH) team to enable them to rescue and assist individuals in the workplace.

The system is based on an Arduino (UNO) platform and is designed to detect hydrogen sulfide using an MQ-136 gas sensor and fires using a flame sensor. Additionally, it utilizes the MQ-2 gas sensor to identify smoke and methane, and the DHT-11 sensor to monitor temperature and humidity levels. When hazardous gases are detected, the sensors become active and transmit this information to the Arduino, triggering an alarm with a buzzer to alert people in the vicinity.



What is it for?

The industrial safety system was created with the primary objective of safeguarding the well-being of individuals working in hazardous environments. It serves to protect against chemical reactions, ensuring the safety of both the workplace and the surrounding environment. Each year, more than 4 million individuals lose their lives due to injuries, with India recording 413,457 accidental injury-related fatalities in 2015. The manufacturing sector is a significant contributor to injury-related illnesses and deaths. This project is designed to reduce the risks faced by industrial workers. If there is a leakage of harmful gases or a potential fire incident, the system issues an alert to workers, enabling them to take life-saving actions. The key benefit of this industrial safety system project is the prevention of accidents and the swift response it facilitates.



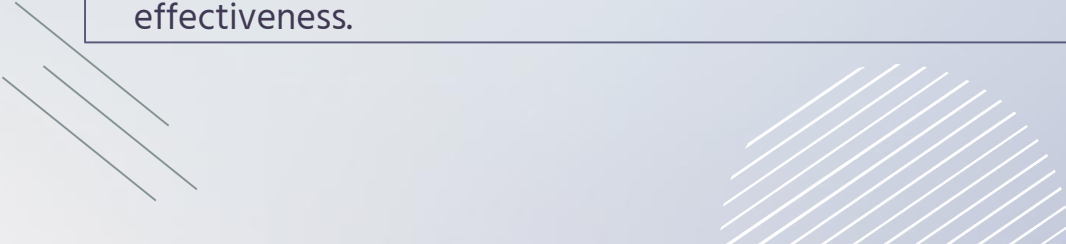


Target Customers

1. Primary End-Users (Workers):

- Needs: Safety and protection from accidents, quick and effective emergency response, ease of use, clear and immediate alerts, proper training and understanding of the system.
- Concerns: Minimizing false alarms, system reliability, comfort in working with the system, user-friendly interfaces.

2. Safety Managers and OSH Teams:

- Needs: Early detection and alerts, centralized monitoring, data analytics for trend analysis, compliance with safety regulations, ease of maintenance, and system diagnostics.
 - Concerns: System accuracy, false positive/negative rates, scalability, cost-effectiveness.
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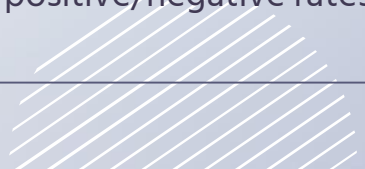
Target Customers

4. Regulatory Authorities:

- Needs: Compliance with safety standards and regulations, access to system data for audit and validation.
- Concerns: System integrity, accurate record-keeping, data privacy and security.

5. Emergency Responders:

- Needs: Accurate and timely information, communication channels, integration with their systems, and coordination tools.
 - Concerns: Data accuracy, real-time information, communication reliability.
- and OSH Teams:

- Needs: Early detection and alerts, centralized monitoring, data analytics for trend analysis, compliance with safety regulations, ease of maintenance, and system diagnostics.
 - Concerns: System accuracy, false positive/negative rates, scalability, cost-effectiveness.
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Hardware Requirements



Arduino Uno

Arduino Uno is a microcontroller from Arduino Family, a friendly board based on the ATmega328

DHT11

The DHT sensor is employed for temperature and humidity measurement. It functions by utilizing a capacitive humidity sensor and a thermistor to gauge the atmospheric conditions.

Flame sensor

it detects the flame and responds to the presence of the flame.

Hardware Requirements

Gas sensor (mq137,mq138)

There are many MQ sensors but the best for alcohol/gas detection is Mq137,138. This sensor provides both digital and analog output.

Buzzer

The buzzer is an electric-sounding device that generates sounds

LED

A light-emitting diode (LED) is a semiconductor light source that emits light when current flows through it.



SOFTWARE REQUIREMENTS

Arduino IDE



The Arduino IDE (Integrated Development Environment) is software employed to upload programs onto boards. Its main purpose is to create projects related to electronics. Arduino is an open-source platform that offers a straightforward and user-friendly coding environment.

Fritzing

Fritzing is an open-source software application designed for electronics enthusiasts, hobbyists, and beginners. It provides a user-friendly platform for designing and documenting electronic circuits and creating PCB (Printed Circuit Board) layouts. Fritzing includes a visual interface where users can drag and drop components to create circuit diagrams, and it also offers tools for designing custom PCBs. It is particularly popular for its educational value and its ability to help users learn and teach electronics and circuit design.

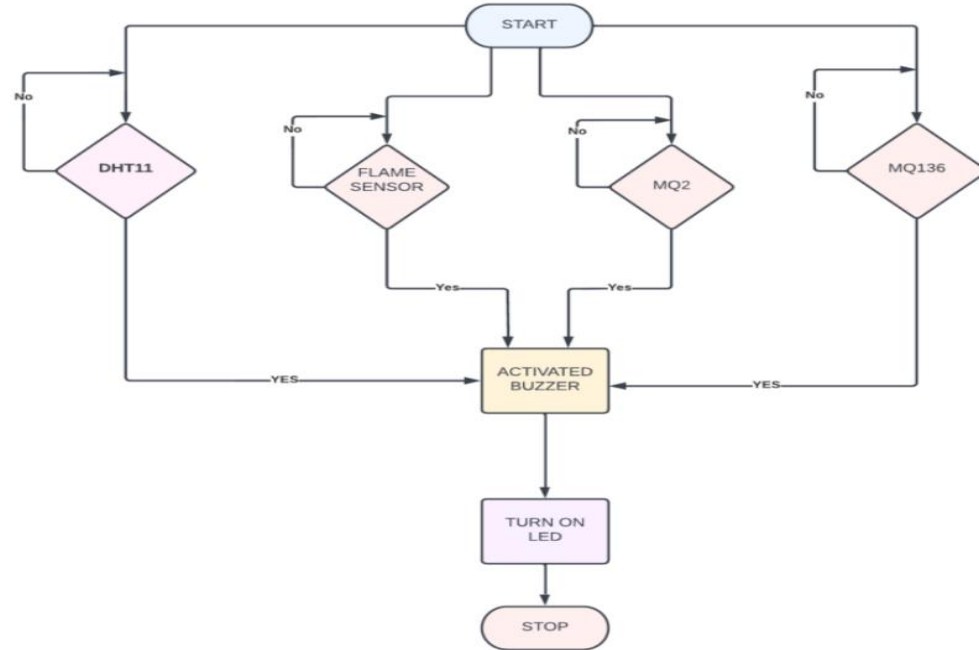


Block Diagram



In the given block diagram, the DHT11, flame sensor, and gas sensor serve as input devices, transmitting data to the Arduino. On the other hand, the buzzer and LED function as output devices, receiving information from the Arduino.

Flow Chart



In a similar vein, the flowchart below outlines the sequential actions within the project and the logical flow of the code. To begin, the flame sensor, DHT11, MQ2, and MQ136 sensors initiate the detection of hazardous substances, excessive humidity, and the presence of fire. If any of these conditions are detected by the sensors, it results in the activation of the LED and buzzer. Conversely, if none of these conditions are met, the buzzer and LED are deactivated.

Pin wiring

Gas sensor (mq137)

The MQ-137 is a gas sensor that has 6 pins :-

1. Heater Power (H)
2. Heater Ground (GND)
3. Analog Output (AOUT)
4. Digital Output (DOUT)
5. Voltage Input (VC)
6. Ground (GND)

Out of these, Ground is connected to the ground of the Arduino board, Voltage input to 5V connection and analog output to A5.

Gas sensor module (mq138)

The MQ-138 is a gas sensor module that has 4 pins :-

1. Heater Control(H1)
2. A1 (Anode connection)
3. B1 (Cathode connection)
4. Ground (GND)

Out of these, Ground is connected to the ground of the Arduino board, A1 is connected to A1 and B1 is connected to A4.



Pin wiring

Flame sensor module (KY-026)

The flame sensor module has 4 pins :-

1. Ground (G)
2. Voltage Input (+)
3. Analog output (A0)
4. Digital output (D0)

All of these are connected to the Arduino board, where A0 is connected to A0, Ground and Voltage Input is connected to Ground and 5V pin respectively and D0 is connected to digital pin 12.

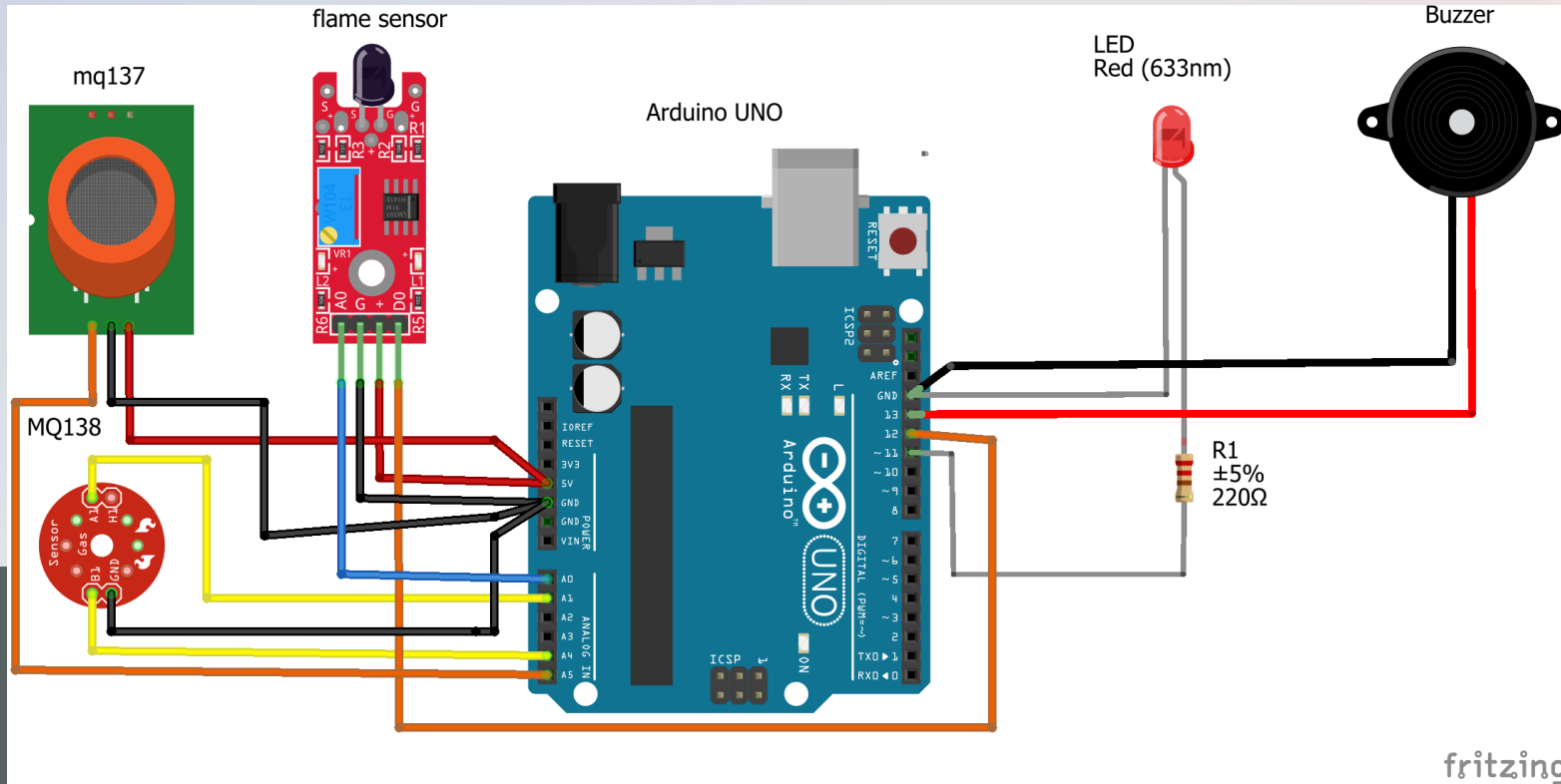
LED and Buzzer

Both of these have a ground pin and a positive (cathode) pin.

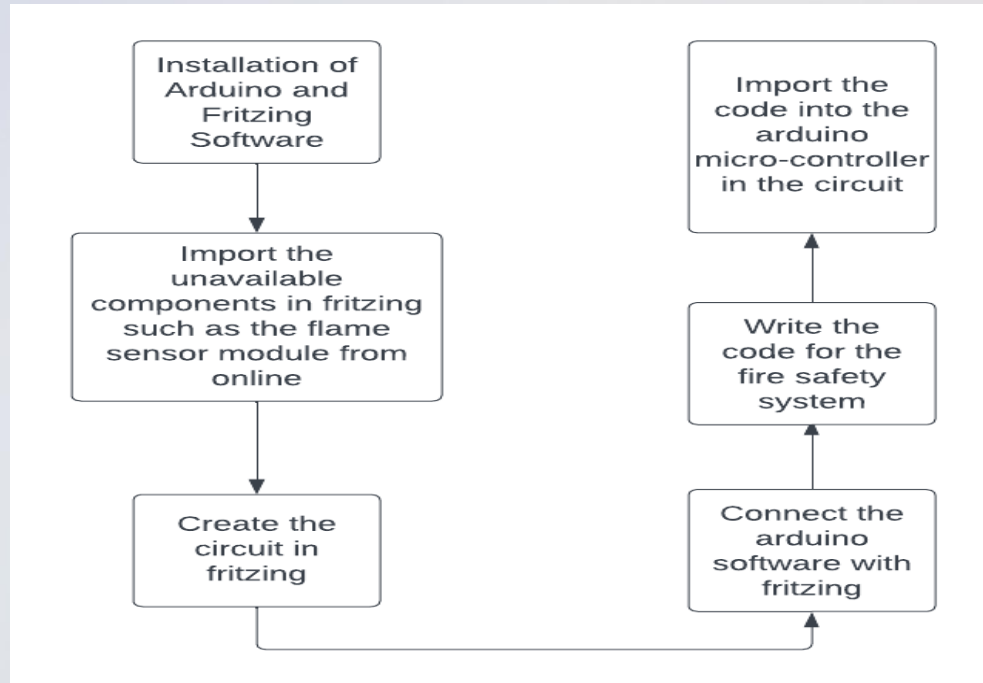
The ground for both is connected to the ground pin of the Arduino board, and the LED positive pin is connected with a resistor to digital pin 11 and the buzzer is connected to digital pin 13.



Circuit Diagram



Software prototype Process Flow





Software prototype Process Flow

1. First download the Arduino and Fritzing software from the official websites and set them up.
2. Next open fritzing and create a new sketch and add the necessary components on the breadboard view.
3. Download the flame sensor module (.fzpz file) from online into the bins folder of Fritzing which was created upon downloading the software.
4. Next go to my parts and import this file and restart the application.
5. Now you can use the flame sensor module; Finish the connections and save it.
6. Next go to code view and write the code for the system.
7. After finishing it, go to preferences > Code view > and import the Arduino application into the fritzing software.
8. Now select the platform, board and port and upload the code to the Arduino board. Now you can run it.
9. Export as PCB to get the gerber files.



Necessary files for the prototype

Arduino Code

<https://drive.google.com/file/d/1Xn4YE34fC7fZnILJEYjPVAgYfojLnycQ/view?usp=sharing>

Gerber files

<https://drive.google.com/drive/folders/1XVMJb-W2Qg-Az6ly-5szzv1imyrxNJRv?usp=sharing>

Video presentation

<https://drive.google.com/file/d/108e5zIG71rTEIMF5L8FYk798EYZWVSsz/view?usp=sharing>



Requirements for Expansion/Future works of the System

- 
1. Distinguishing between an electrical fire and a standard fire.
 2. Provision of water for extinguishing regular fires.
 3. Availability of sand for addressing electrical fires.
 4. Establishing connections with emergency services for swift responses.
 5. Backup power supply for the system in case of power failure.
 6. Detection capabilities for pinpointing the origin of a fire.
 7. More efficient and compact system for larger scale.
- 



CONCLUSION

In conclusion, the industrial safety system project represents a vital step towards safeguarding human lives and valuable assets in industrial settings. With the integration of various sensors like the MQ-137, flame sensor module, MQ-138, and DHT-11, this system provides comprehensive coverage for detecting and responding to potential hazards, including gas leaks and fire outbreaks.

Therefore, we initiated the industrial safety system project, emphasizing the paramount significance of prioritizing the well-being of workers. These factors are integral in product development, as they contribute to enhanced production levels.

